A sharp rank-eight extremal marginal state in dimension six

Candidate partial solution packet for arXiv:2412.10041

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Status. Candidate substantial partial solution, computer-assisted but exact. We settle the first equal-dimension case omitted by the source: $\mathcal{MR}(I_6/6, I_6/6) = 8$. The general construction problem remains open.

1 Source question

For states $\rho_1 \in \mathbb{M}_{d_1}$ and $\rho_2 \in \mathbb{M}_{d_2}$, let $\mathcal{C}(\rho_1, \rho_2)$ be the convex set of bipartite states with these two marginals. Parthasarathy's bound says that every extreme $\rho \in \mathcal{C}(\rho_1, \rho_2)$ satisfies

$$\operatorname{rank} \rho \leq \left\lfloor \sqrt{d_1^2 + d_2^2 - 1} \right\rfloor .$$

Devendra, Dey, and Dey ask for constructions attaining this bound. Their paper handles $d_1 = d_2 = 3, 4, 5, 9, 12$ and certain multiples of five, but not $d_1 = d_2 = 6$. In that case the bound is

$$\left\lfloor \sqrt{2 \cdot 6^2 - 1} \right\rfloor = \lfloor \sqrt{71} \rfloor = 8.$$

2 Proof intuition

Under the Choi–Jamiołkowski correspondence, it is enough to construct a bistochastic completely positive map on $\mathbb{M}_6$ with eight Kraus operators. The marginal constraints ask for

$$\sum_j U_j^* U_j = I_6 = \sum_j U_j U_j^* .$$

Extremality asks that the 64 ordered pairs

$$\left(U_i^* U_j,\ U_j U_i^*\right), \qquad 1 \leq i, j \leq 8,$$

be linearly independent in $\mathbb{M}_6 \oplus \mathbb{M}_6$.

The construction makes every U_j monomial. Across every input column, and also across every output row, the eight nonzero moduli are exactly $1, 2, \ldots, 8$. Both marginal identities then follow from the same one-line sum of squares. The remaining independence question is finite linear algebra over the Gaussian integers; one nonzero determinant modulo a prime is an exact certificate.

3 The construction

Let $e_1, \dots, e_6$ be the standard basis of $\mathbb{C}^6$. Define eight matrices by listing their columns:

$$
\begin{aligned}
W_1 &= [-8e_3, -8ie_4, -8ie_5, 6ie_2, e_1, 4e_6],\\
W_2 &= [7e_1, -7e_5, -3e_2, -5e_6, -6e_4, -7ie_3],\\
W_3 &= [4e_3, 2e_1, 2e_6, -7e_2, 7ie_4, 2ie_5],\\
W_4 &= [-3e_4, 3e_5, 5ie_3, 4e_1, -8ie_6, 8ie_2],\\
W_5 &= [2e_4, 6ie_6, 4ie_2, -8ie_1, 4ie_5, ie_3],\\
W_6 &= [6e_1, 5e_4, 6e_3, -e_6, 5ie_5, 5ie_2],\\
W_7 &= [5ie_1, -e_2, -ie_4, 2ie_3, 3ie_6, -6e_5],\\
W_8 &= [-e_5, -4ie_4, 7e_6, -3e_1, -2e_2, 3ie_3].
\end{aligned}
$$

Lemma 1. *The matrices above satisfy*

$$\sum_{j=1}^{8} W_j^* W_j = 204 I_6 \quad \textit{and} \quad \sum_{j=1}^{8} W_j W_j^* = 204 I_6.$$

Moreover, the 64 *pairs*

$$\left\{ \left(W_i^* W_j, W_j W_i^* \right) : 1 \le i, j \le 8 \right\}$$

are linearly independent over $\mathbb{C}$.

Proof. Every W_j is monomial. Inspection of the displayed columns shows that, for each fixed input coordinate, the eight nonzero moduli form the multiset $\{1, \dots, 8\}$. After transporting each entry to its output row, the same is true for every fixed output coordinate. Off-diagonal entries in both sums vanish, while each diagonal entry equals

$$1^2 + 2^2 + \cdots + 8^2 = 204.$$

For independence, form a 72×64 Gaussian-integer matrix A. Its columns are indexed lexicographically by (i, j). The first 36 rows are the entries of $W_i^* W_j$, in row-major order, and the final 36 rows are the entries of $W_j W_i^*$, again in row-major order. Let D be the leading 64×64 minor of A; thus D uses all first-product rows and the first 28 second-product rows.

Reduce $\mathbb{Z}[i]$ to $\mathbb{F}_{101}$ by sending i to 10. This is a ring homomorphism because $10^2 \equiv -1 \pmod{101}$. Exact Gaussian elimination gives

$$\det D \equiv 19 \pmod{101}.$$

Hence $\det D \neq 0$ in $\mathbb{Z}[i]$, so A has column rank 64 over $\mathbb{C}$. The claimed pairs are therefore linearly independent. The included verifier reconstructs all matrices and performs both normalization checks and this modular determinant computation using integer arithmetic only. $\square$

Theorem 2. *There is an extreme point*

$$\rho \in \mathcal{C}(I_6/6, I_6/6)$$

of rank 8*. Consequently,*

$$\boxed{\mathcal{MR}(I_6/6, I_6/6) = 8.}$$

Proof. Put $U_j = W_j / \sqrt{204}$ and define

$$\Phi(X) = \sum_{j=1}^{8} U_j^* X U_j, \qquad X \in \mathbb{M}_6.$$

The two identities in Lemma 1 say that Φ is unital and trace preserving. The extremality criterion recalled as Theorem 2.12 in the source says that Φ is extreme among such maps exactly when the pairs $(U_i^*U_j, U_jU_i^*)$ are linearly independent. Lemma 1 gives this independence.

It also forces the U_j's themselves to be linearly independent. Indeed, if $\sum_j c_jU_j = 0$, fixing a nonzero U_k yields simultaneous relations

$$\sum_j c_jU_k^*U_j = 0, \qquad \sum_j c_jU_jU_k^* = 0,$$

and pairwise independence gives every $c_j = 0$. Thus the Kraus decomposition is minimal and the Choi rank of Φ is 8.

Now set $\rho = \mathcal{J}(\Phi)/6$. The Choi correspondence gives $\rho \in \mathcal{C}(I_6/6, I_6/6)$, preserves extremality, and identifies $\operatorname{rank}\rho$ with the Choi rank. Hence ρ is extreme of rank 8. Parthasarathy's upper bound is also 8, proving equality. $\square$

4 Exact verification

Run

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python verify_exact.py
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from the packet directory. The script uses no numerical linear algebra or external package. It verifies the two identities with $204I_6$, builds the 64×64 minor, and returns determinant 19 in $\mathbb{F}_{101}$. The certificate is deliberately stronger than a floating-point rank computation.

5 Upgrade attempts, limitations, and novelty

The first construction attempt used arbitrary Kraus matrices and operator scaling, but it would have left irrational numerical normalization without a compact certificate. The successful upgrade replaced it by monomial Gaussian-integer matrices whose moduli form two simultaneous edge colourings; this makes both marginals exact before the rank test.

A second upgrade generalized the search to other omitted dimensions. Exact full-rank modular candidates appeared immediately in several cases, strongly suggesting a generic phenomenon. They are not claimed here because a uniform proof that the relevant determinant polynomial is nonzero for every dimension is still missing. A dimension-count/transversality argument alone is insufficient: the $d = 2$ bistochastic case is a genuine exception despite the naive count. Likewise, tensor-product propagation gives only finite ranges because Choi ranks multiply while the sharp integer bound follows $\lfloor\sqrt{2}\,d\rfloor$.

Thus Theorem 2 completely settles the first omitted dimension but not the source's general algorithmic question.

A bounded primary-source search through 12 August 2026 used the exact source title and arXiv id, the phrase extremal marginal state dimension 6 rank 8, the phrase bistochastic channel Choi rank 8, and the Parthasarathy sharpness question. It found the source's cases and older general extremality criteria, but no construction for $d = 6$ and no later general resolution. Novelty confidence is moderate; specialist review is recommended before any priority claim.

References

[1] R. Devendra, P. Dey, and S. Dey, *On the rank of extremal marginal states*, arXiv:2412.10041 (2024).

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Lemma 5.7. *Given $d \geq 2$. Let $\{W_j\}_{j=1}^{d+1} \subseteq \mathbb{M}_{d\times(d+1)}$ be the set of matrices given by*

$$\begin{cases} W_1 = [e_1^{(d)}, e_2^{(d)}, \cdots, e_d^{(d)}, 0], \\ W_2 = [0, e_1^{(d)}, e_2^{(d)}, \cdots, e_d^{(d)}], \\ W_3 = [e_d^{(d)}, 0, e_1^{(d)}, \cdots, e_{(d-1)}^{(d)}], \\ \vdots \\ W_{d+1} = [e_2^{(d)}, e_3^{(d)}, \cdots, e_d^{(d)}, 0, e_1^{(d)}]. \end{cases} \tag{5.4}$$

Then the pair of sets $\{W_i^*W_j : 1 \leq i, j \leq d+1\} \subseteq \mathbb{M}_{d+1}$ and $\{W_jW_i^* : 1 \leq i, j \leq d+1\} \subseteq \mathbb{M}_d$ is bi-linearly independent.

Proof. The proof is in similar lines to the proof of the Lemma 5.4. □

Theorem 5.8. *Given $d \geq 2$. There exists an extreme point of the set $\mathcal{CP}(\mathbb{M}_d, \mathbb{M}_{d+1}; \frac{I_d}{d}, \frac{I_{d+1}}{d+1})$ with Choi-rank $d+1$.*

Proof. Let $\Phi : \mathbb{M}_d \to \mathbb{M}_{d+1}$ be the map defined by $\Phi = \frac{1}{d^2+d}\sum_{j=1}^{d+1} \mathrm{Ad}_{W_j}$, where W_j's are given by (5.4) for $1 \leq j \leq d+1$. Check that $\Phi \in \mathcal{CP}(\mathbb{M}_d, \mathbb{M}_{d+1}; \frac{I_d}{d}, \frac{I_{d+1}}{d+1})$. By Lemma 5.7, it follows that the pair of sets $\{W_i^*W_j\}_{i,j=1}^{d+1} \subseteq \mathbb{M}_{d+1}$ and $\{W_jW_i\}_{i,j=1}^{d+1} \subseteq \mathbb{M}_d$ is bi-linearly independent. Hence, by Theorem 2.12, we conclude that Φ is an extreme point of the set $\mathcal{CP}(\mathbb{M}_d, \mathbb{M}_{d+1}; \frac{I_d}{d}, \frac{I_{d+1}}{d+1})$ with $CR(\Phi) = d+1$. □

Corollary 5.9. *If $d \geq 2$, then $\mathcal{MR}(\frac{I_d}{d}, \frac{I_{d+1}}{d+1}) \geq d+1$. Moreover, $\mathcal{MR}(\frac{I_d}{d}, \frac{I_{d+1}}{d+1}) = d+1$ if $d = 2, 3$.*

Proof. The first part is immediate from the Theorem 5.8. The second part also follows because $\mathcal{MR}(\frac{I_d}{d}, \frac{I_{d+1}}{d+1}) \leq \lfloor\sqrt{d^2+(d+1)^2-1}\rfloor = d+1$ if $d = 2, 3$. □

We wish to conclude this paper with the following mentioned. Given $d_1 = d_2 \geq 3$ or $d_1 \neq d_2$ with $d_1, d_2 \geq 2$, this would be interesting to study whether it is possible to give an algorithm to construct an extreme point that attain the upper bound given by K. R. Parthasarathy. We are not able to answer this at this stage. We leave this for furthur investigation in the future.

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7. Appendix

Lemma 7.1. *Let $\{W_j : 1 \leq j \leq 6\} \subseteq \mathbb{M}_5$ be the set of matrices given by*

$$\begin{cases} W_1 &= E_{13} + E_{32} = [0, e_3, e_1, 0, 0], \\ W_2 &= E_{24} + E_{43} = [0, 0, e_4, e_2, 0], \\ W_3 &= \sqrt{2}E_{35} + E_{54} = [0, 0, 0, e_5, \sqrt{2}e_3], \\ W_4 &= E_{14} + E_{42} = [0, e_4, 0, e_1, 0], \\ W_5 &= E_{15} + E_{41} + E_{53} = [e_4, 0, e_5, 0, e_1], \\ W_6 &= \sqrt{2}E_{21} + E_{52} = [\sqrt{2}e_2, e_5, 0, 0, 0]. \end{cases} \tag{7.1}$$

Then the pair of sets $\{W_i^*W_j : 1 \leq i, j \leq 6\}$ and $\{W_jW_i^* : 1 \leq i, j \leq 6\}$ is bi-linearly independent.

Figure 1: The source's concluding construction problem, on PDF page 17.